

LEMON: Factor-Level Semantic Alignment for Graph–Text Pairs

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Abstract. Graph–text evaluation often misses a central failure mode: a sentence may preserve the entity names while changing or omitting the relation encoded by the graph edge. We introduce LEMON (Linguistically Enriched Measure of Ontological Normalization), a diagnostic metric for bidirectional knowledge-graph (KG)–text fidelity. LEMON decomposes each graph predicate into weighted semantic factors and evaluates both KG→Text coverage and Text→KG recoverability. Its goal is not to be the most severe scalar detector. Instead, it explains relation-level errors by localizing them to argument roles, relation class, direction, polarity, or textual evidence. On WebNLG, Forward LEMON reaches 0.792 and Reverse LEMON reaches 0.779. A MINE-style comparison confirms that node information is easier to recover than edge information, while worked examples and controlled perturbations show why a factor trace is more informative than a single edge-failure label. The same inventory format is applied to DrugProt and BC5CDR, where biomedical predicates make role, direction, polarity, and causality explicit. The inventories and weights are auditable design resources, not yet expert-certified semantic gold standards. Code and reproducibility materials are available at <https://github.com/Cogniscy/lemon>.

Keywords: Graph-to-text evaluation · Semantic alignment · Knowledge graphs · WebNLG · Biomedical relation extraction · Predicate decomposition

1 Introduction

Although a graph edge is compact, it encodes more than the names of two entities. In an edge (s, p, o) , the predicate p fixes the relation between the arguments: who did what, to whom, in which direction, and with which semantic commitment. A graph-to-text system can mention the entities and still lose the predicate. The reverse case is similar: a reconstructed graph is useful only if it recovers the relation, not merely the endpoints.

This distinction matters in ordinary examples. A sentence can mention *Alan Bean* and *Wheeler, Texas* without saying that Wheeler is his place of birth. A biomedical abstract can keep the same chemical and protein names while turning inhibition into activation. Standard graph-to-text evaluation usually starts

from reference similarity: BLEU counts overlapping word n-grams, METEOR aligns words, stems, and synonyms, chrF++ measures character and word n-gram F-scores, TER estimates the edits needed to reach a reference, BERTScore compares contextual token embeddings, and BLEURT learns a reference-based quality signal [5–10]. These metrics are useful for generation quality, but they rarely identify which part of a source predicate was preserved, weakened, or lost.

LEMON treats the predicate as the unit of diagnosis. Each predicate is decomposed into typed, weighted semantic factors. For example, *birthPlace* contains a person-like subject, a biographical relation, and a place-like object. A DrugProt inhibition predicate contains a chemical actor, a gene/protein target, negative regulation, directionality, and evidence form. The score evaluates whether these factors are expressed in text or recovered in a reconstructed graph, and it records which factors are missing.

The method is deliberately narrower than general semantic similarity. The source graph supplies the semantic structure, and LEMON checks whether the relevant predicate factors survive across graph and text. This makes the metric suitable for graph–text fidelity settings, including RDF-to-text evaluation and relation-centered biomedical transfer, but not for open-ended text–text similarity.

The closest comparison point in this paper is MINE-style scoring, inspired by KGen’s Measure of Information in Nodes and Edges (MINE) [22]. A node/edge score is closer to graph fidelity than a sentence metric because it separates recovered entities from recovered relations. Its limitation in our setting is resolution: missing relation class, reversed roles, and wrong polarity all collapse into a similar edge-level failure. LEMON keeps the graph-aware view but opens the edge into weighted semantic factors.

The experiments follow this argument across three resources. WebNLG is a Resource Description Framework (RDF)-to-text benchmark built from DBpedia triples; it lets us test whether entity overlap hides relation-level errors [1]. DrugProt is a biomedical corpus of chemical–gene/protein relations, and BC5CDR is a corpus of chemical-induced disease relations [20, 21]. We first evaluate WebNLG in both KG→Text and Text→KG directions, then stress metrics with controlled perturbations, transfer the inventory format to the biomedical corpora, and run a small three-judge LLM probe to test factor-level recoverability.

The contributions are: (i) a bidirectional factor-level metric for KG–text fidelity; (ii) fixed factor inventories for WebNLG, DrugProt, and BC5CDR; (iii) reproducible forward, reverse, perturbation, component, and LLM-judge analyses; and (iv) a direct comparison with coarse node/edge and triple-style views. The paper is diagnostic rather than leaderboard-oriented: it shows not only that a relation failed, but whether the failure concerns role assignment, direction, polarity, relation class, or evidence.

2 Related Work

2.1 RDF-to-text evaluation and reference metrics

The WebNLG Challenge made Resource Description Framework (RDF)-to-text generation a standard test bed: systems receive DBpedia-derived triples and produce natural-language verbalizations [1]. WebNLG+ later added bilingual and bidirectional settings, so the same data family can support graph-to-text generation and text-to-graph recovery [2]. GEM placed such tasks within a broader natural language generation (NLG) evaluation suite, and recent LLM-based graph-to-text work, such as PlanGTG, revisits planning and attribution in graph-conditioned generation [3, 4].

These benchmarks often report sentence-level reference metrics. BLEU measures word n-gram precision; METEOR aligns words, stems, and synonyms while balancing precision and recall; chrF++ uses character n-gram F-score with a word n-gram extension; TER counts edits needed to transform an output into a reference [5–8]. BERTScore compares contextual token embeddings, and BLEURT learns a reference-based quality signal from human-rated examples [9, 10]. These metrics are useful for reference similarity and generation quality. They do not identify whether a source predicate preserved its argument roles, direction, polarity, or evidence.

2.2 Source-aware factuality and graph metrics

Source-aware metrics move closer to the input structure. PARENT weights n-gram precision and recall by evidence in the source table and reference text; Data-QuestEval casts data-to-text coverage as question answering; AlignScore learns a general alignment function for factual consistency [11–13]. FactSpotter is the closest prior graph-to-text evaluation approach: it checks whether a triple (s, p, o) is present in a generated text and correlates with human judgments of correctness, coverage, and relevance [14]. LEMON differs by replacing a binary whole-triple decision with weighted predicate-factor diagnosis.

Text-driven factuality metrics take the opposite route. They decompose generated text into verifiable claims and check support against evidence, as in FActScore and VERISCORE [15, 16]. Semantic graph metrics compare structures under node alignment, as in Smatch, S²match, and Smatch++ for Abstract Meaning Representation (AMR) parsing [17–19]. LEMON is source-driven: it decomposes the graph predicate first and evaluates the text against that decomposition.

2.3 MINE-style recovery and LLM judging

Text-to-knowledge-graph (text-to-KG) evaluation asks whether extracted graphs retain information from text. KGGen introduces a language-model-based extractor and MINE, the Measure of Information in Nodes and Edges, for node and edge retention [22]. Our MINE-style baseline adapts this node/edge split to

WebNLG graph-text pairs. It is a useful coarse graph-fidelity signal, but it does not separate polarity errors, role swaps, and missing relation class; all can appear as edge-level failure.

LLM-based evaluation can recover meaning that lexical rules miss, but it must be separated from reproducible scoring. G-Eval and MT-Bench illustrate both the scale of LLM judging and its calibration risks [26, 27]. Related retrieval work uses LLMs to refine ambiguous relevance judgments, including hard-negative filtering when raw similarity scores are insufficient [28]. We use LLMs only in an exploratory factor-judging probe; the main tables are rule-based and reproducible without API calls.

2.4 Behavioral metric calibration

A metric should react predictably to controlled changes. CheckList introduced invariance and directional tests for NLP models [23]. Perturbation CheckLists and synthetic stress tests apply the same idea to NLG evaluation [24, 25]. Our perturbations follow this logic: relation deletion, argument swap, polarity flip, and relation blur test whether a metric reacts to intended semantic damage.

3 Method

3.1 Problem setting

Let a graph-text pair be (G, T) , where G is a set of directed edges $e = (s, p, o)$ and T is a verbalization. The forward task evaluates whether the text covers the information in the graph. The reverse task evaluates whether a graph reconstructed from the text, denoted $\hat{G}(T)$, preserves the information in the original graph. LEMON uses the same predicate-level representation in both directions.

3.2 Predicate factors and weights

Each predicate p is mapped to a weighted decomposition:

$$D(p) = \{(f_i, r_i, w_i)\}_{i=1}^n, \quad \sum_i w_i = 1,$$

where f_i is a semantic factor, r_i is its role, and $w_i > 0$ is a salience weight. Roles cover predicate meaning, subject and object domains, value domains, and modifiers such as direction, polarity, causality, or evidence. The representation is coarser than a full ontology but richer than a single predicate label. A text that keeps both entities but loses the relation cue can therefore receive partial credit rather than full coverage.

The inventories are fixed before scoring. They are manually specified diagnostic resources built from predicate labels, argument roles, domain constraints, entity types, and lexical evidence cues. The weights are normalized salience priors. Predicate-meaning factors usually receive the largest share; domain factors

Coarse mismatch vs. factor-level diagnosis

$$\text{LEMON}(e, T) = \sum_i w_i \cdot m_i$$

w_i = factor weight m_i = evidence match in text (0, 0.5, 1)
 ✓ preserved ½ partial × missing/contradicted

A. Polarity error

Triple: Aspirin — inhibits — COX-1

Text: "Aspirin activates COX-1."

MINE-style

nodes: Aspirin ✓, COX-1 ✓

edge/fact: inhibits ×

score ≈ 0.50

LEMON-Factor

factor	weight×match	contrib.
chemical actor	0.20 × 1.0	0.20
protein target	0.20 × 1.0	0.20
regulation relation	0.15 × 0.5	0.075
negative polarity	0.30 × 0.0	0.000
direction chem→protein	0.15 × 1.0	0.150

LEMON score = 0.625

0.20	0.20	0.075	0.000	0.150
chemical actor	protein target	regulation relation	negative polarity	direction chem→protein

Diagnosis: polarity contradicted

B. Relation-class loss

Triple: Alan Bean — birthPlace — Wheeler, Texas

Text: "Alan Bean lived in Wheeler, Texas."

MINE-style

nodes: Alan Bean ✓

Wheeler ✓

edge/fact: birthPlace ×

score ≈ 0.50

LEMON-Factor

factor	weight×match	contrib.
person subject	0.25 × 1.0	0.25
place object	0.25 × 1.0	0.25
birth relation	0.35 × 0.0	0.00
direction person→place	0.15 × 1.0	0.15

LEMON score = 0.65

0.25	0.25	0.00	0.15
person subject	place object	birth relation	direction person→place

Diagnosis: birth relation missing

Same coarse failure, different semantic diagnosis: MINE-style matching collapses both cases into relation mismatch, while LEMON-Factor preserves graded partial credit and localizes the error.

Fig. 1. Worked examples of factor-level scoring. The displayed weights come from the current predicate inventories and are normalized to sum to one. The MINE-style view collapses both cases into a coarse relation mismatch, whereas LEMON assigns partial credit to preserved factors and localizes the failure to polarity or relation class. The examples illustrate scoring mechanics, not a separate benchmark.

and modifiers receive smaller shares when they are part of the predicate meaning. These weights are auditable design choices, not learned parameters or expert-certified semantic constants. An expert validation protocol has been prepared, but expert results are not included in this version.

Figure 1 shows two worked examples. MINE-style collapses both cases into a coarse relation mismatch. LEMON preserves partial credit for the factors that remain correct and localizes the error to polarity or relation class.

3.3 Forward LEMON: KG → Text coverage

For an edge e , text T , and component (f_i, r_i, w_i) , each factor is checked by an evidence function

$$E(f_i, r_i, e, T) \in [0, 1].$$

In the main reproducible experiments, E is implemented with fixed lexical and domain rules: entity and domain factors use normalized label or token evidence, while relation factors use predicate cues and factor cues. The edge-level forward score is:

$$S_F(e, T) = \sum_i w_i E(f_i, r_i, e, T),$$

because the weights are normalized. Example-level and corpus-level scores are averages over edges with available decompositions. The implementation stores covered and missing factors for each edge, so the scalar score is paired with a diagnostic trace.

3.4 Reverse LEMON: Text→KG recoverability

The reverse direction first constructs a rule-based reconstructed graph $\hat{G}(T)$ from the text using entity-label evidence and predicate lexical cues. Reverse LEMON then compares each gold edge with its best recovered counterpart. Subject and object recovery contribute through node/domain evidence; predicate recovery contributes through overlap between the gold predicate’s factors and the recovered edge evidence. The reverse score evaluates whether the source predicate’s semantic factors survive text-to-graph reconstruction.

The current reconstructed graph is a lexical pseudo-extractor, not a full KG extraction model. This choice keeps the experiment reproducible and keeps the focus on metric behavior rather than extractor quality. Future work can replace $\hat{G}(T)$ with KGGen, GraphRAG-style extraction, OpenIE, or domain-specific extractors.

3.5 MINE-style node/edge baseline

To compare factor-level recoverability with node/edge information retention, we implement a MINE-style baseline inspired by KGGen’s MINE. Each gold triple is treated as a fact, and the reconstructed graph is searched for matching or related nodes and edges. The score separates node information, edge information, and a composite partial-credit node/edge score. We report it as a local baseline rather than a reproduction of the original benchmark.

3.6 Controlled perturbation calibration

A graph-text metric should react to meaning-destroying perturbations and remain comparatively stable under meaning-preserving perturbations. We create auditable text and graph perturbations while keeping the source graph fixed. Destructive operations include relation phrase deletion, object swaps, predicate swaps, node drops, and edge drops. Preserving operations include alias-like rewrites and template paraphrases. Weak-template cases are marked when a template preserves entity strings but may alter phrasing in a way that is less reliable. The perturbation files and operation manifests are saved with the repository.

4 Data and Experimental Resources

4.1 WebNLG sample

The WebNLG experiments use a stratified English sample from the RDF-to-text benchmark introduced above. Each item pairs a small DBpedia-derived graph

with one or more English verbalizations. The train split contains 200 examples and the development split contains 100 examples; both cover 16 WebNLG categories. The development set contains 308 graph edges. The sample is small by design: it keeps the scoring pipeline auditable while still exposing relation-level errors that entity overlap can hide.

4.2 Biomedical transfer resources

The biomedical transfer checks use two relation extraction corpora converted into the same graph-text JSONL format. DrugProt is a BioCreative VII resource for chemical-gene/protein relations with 13 relation types and 24,526 manually annotated relations [20]. BC5CDR is a BioCreative V collection of 1500 PubMed articles annotated for chemicals, diseases, and chemical-induced disease (CID) relations; it contains 3116 chemical-disease interactions [21]. In our processed subsets, DrugProt provides 500 train abstracts with 3588 edges, and BC5CDR provides 500 train and 500 development documents with 985 and 961 CID edges.

Biomedical text stresses the dimensions that motivate LEMON. In scientific abstracts, a change from inhibition to activation, a reversed chemical-protein relation, or a weakened causal claim can alter the interpretation of the result. Literal evidence such as dosages, measurements, and experimental conditions also becomes important when text is translated into a graph. DrugProt and BC5CDR are therefore used as diagnostic transfer settings: they test role, direction, polarity, causality, and evidence under domain conditions where surface entity overlap is not enough. We do not train or claim a new biomedical relation extraction system.

4.3 Predicate-factor inventories

The experiments use fixed JSON predicate-factor inventories. Each inventory contains a controlled factor vocabulary and one weighted decomposition per predicate. WebNLG uses 121 predicate decompositions and covers all 119 predicate labels observed in the evaluation files. DrugProt contains 13 biomedical relation decompositions and covers the 12 relation types observed in the processed subset. BC5CDR is intentionally a single-relation setting: its inventory decomposes the document-level chemical-induced disease relation into chemical exposure, disease outcome, causality, document-level evidence, and chemical-to-disease direction.

The inventories are rule-based diagnostic resources. Earlier LLM-generated decomposition candidates are kept as development material, but they are not treated as expert gold and are not used as the main evidence in the scoring tables below. The validator checks that coverage is complete for the current evaluation files and that each predicate decomposition has normalized weights.

5 Experiments

The experiments test one representation from several angles. WebNLG provides an explicit RDF-to-text setting. DrugProt and BC5CDR test whether the same factor format remains meaningful when predicates encode biomedical roles, direction, polarity, and causality. Main scoring uses JSONL inputs and fixed JSON factor inventories. LLM calls appear only in the reliability probe; they are not used to construct the rule-based tables.

5.1 Forward and reverse factor scoring

The first WebNLG experiment scores development texts using predicate decompositions and lexical cues. We run an error-driven resource expansion: missing predicates are identified, rule-based decompositions are added, lexical cues are expanded, and coverage is recomputed. This separates missing evaluation resources from missing information in the text.

To test bidirectionality, we reconstruct a graph from each WebNLG text with a lexical pseudo-extractor and compute Reverse LEMON. The reverse score is compared with expanded Forward LEMON. A small forward–reverse gap would indicate that the same factor schema can support both $\text{KG} \rightarrow \text{Text}$ coverage and $\text{Text} \rightarrow \text{KG}$ recoverability.

5.2 Node/edge retention and perturbations

We compare Reverse LEMON with MINE-style, which reports node information, edge information, and a composite node/edge score. A locked 50-fact subset shows how binary fact recoverability differs from partial-credit node/edge retention. These values are diagnostic baselines, not a reproduction of the full KGGen/MINE benchmark.

Controlled perturbations are injected into text and reconstructed graphs. The source graph remains fixed, so the test measures whether a metric reacts to intended semantic damage in the text. The final scoring uses five destructive operators: node deletion, relation-cue deletion, argument swap, polarity flip, and relation blur. Polarity flips are most informative for DrugProt and BC5CDR, where relation labels explicitly encode activation, inhibition, regulation, or causality.

We score the shared perturbation files with entity recall, label match, binary triple match, a MINE-style node/edge harmonic score, a label-only LEMON variant, and the full LEMON score based on weighted predicate factors. The same scoring layer is used for the component analysis. An offline character n-gram vector baseline is reported as a lexical/topical control, not as a dense embedding benchmark.

5.3 Exploratory LLM factor judging

The LLM reliability probe evaluates semantic recoverability. We sample perturbed instances from WebNLG, DrugProt, and BC5CDR and ask three independent LLM judges to label each semantic factor as covered, partially covered,

or absent. The judges are Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B, called through OpenRouter with JSON-schema constrained outputs. This probe instantiates the evidence decision with LLMs to test stability, but it is not human validation and it is not used in the main rule-based scores.

6 Results

6.1 From entity overlap to relation factors

The first WebNLG experiment shows why predicate resources must be explicit. With the initial factor dictionary, only 140 of 308 development edges could be scored, and the missing decomposition rate was 0.546. After rule-based decomposition and cue expansion, all 308 edges had predicate factors and Forward LEMON rose from 0.368 to 0.792. This increase does not mean that all WebNLG texts are faithful. It shows that relation-level diagnosis requires predicate resources before text quality can be evaluated.

Lexical controls identify the added signal. Exact entity-label overlap is already high at 0.922 and does not change under predicate-resource expansion. Token cosine and Jaccard remain fixed because the graph-text strings do not change. LEMON changes because it checks relation roles, predicate meanings, and lexical evidence instead of counting entity mentions alone.

The reverse run checks the same representation from text back to graph. A lexical pseudo-extractor recovered 390 of 408 nodes and 196 of 308 edges from the WebNLG texts. Reverse LEMON reached 0.779, close to the expanded Forward score of 0.792. The MINE-style comparison explains the remaining difficulty: its composite score is 0.803, but the locked 50-fact subset has node information of 0.960 and edge/fact information of only 0.540. The relation is harder to recover than the names of the entities.

A relation-deletion sanity check isolates the same failure mode. Removing relation evidence changes exact entity-label coverage by only 0.002, while Forward LEMON drops by 0.211. Predicate cue coverage and MINE fact recoverability fall by 0.364 and 0.354. Preserving entity names is therefore not the same as preserving the predicate.

6.2 Cross-domain perturbation sensitivity

The cross-domain perturbation files test which metrics react when controlled edits damage relation meaning in WebNLG, DrugProt, and BC5CDR. The current inventories audit entity/domain support, predicate cues, participant roles, direction, polarity, and textual evidence. A perturbation can be diagnosed only when the corresponding factor group is present in the inventory.

Table 1 reports normalized score loss under five perturbations. The columns are different preservation signals rather than one shared accuracy scale: entity recall tracks entity survival; triple match and MINE-style node/edge scoring act as coarse edge/fact alarms; LEMON label-only removes factor decomposition; LEMON is the full weighted factor score.

Table 1. Perturbation sensitivity as normalized score loss (%). Entity recall checks source entity mentions; Triple match checks whole SPO recovery; MINE-style is the adapted node/edge baseline; LEMON label-only removes factor decomposition; LEMON is the full weighted factor score.

Perturbation	Entity recall	Triple match	MINE-style node/edge	LEMON label-only	LEMON
Node deletion	32.6	67.5	65.7	32.3	35.2
Edge deletion	21.5	65.5	63.7	48.2	28.1
Argument swap	21.4	54.0	52.1	32.1	41.3
Polarity flip	21.5	54.6	52.6	33.7	41.4
Relation blur	21.6	56.8	54.9	38.5	28.1

Table 2. Component sensitivity analysis for LEMON. Loss is normalized score loss under perturbation, reported in percent. Gain is the loss of full LEMON minus the loss after removing a component; positive gain means that the removed component contributed to sensitivity. This table explains LEMON internals rather than ranking LEMON against other metrics.

Variant	Loss (%)	Gain vs. full (pp)
LEMON	32.0	0.0
Label-only	41.7	-9.7
Unweighted	31.0	1.0
Minus roles	18.5	13.5
Minus direction	31.0	1.0
Minus polarity	30.4	1.6
Minus evidence	29.0	3.0

The table should not be read as a global ranking. Triple match and MINE-style scoring often react more strongly because they treat edge damage as a coarse failure. LEMON is most useful when the reason for failure matters. It reacts to argument swaps and polarity flips when those dimensions are encoded, and it links the response to factor groups that can be inspected.

Table 2 is an internal component analysis, not a comparison with MINE-style or triple matching. Removing participant-role factors produces the largest positive gain for the full model over the ablated variant, so role factors are the largest observed source of diagnostic sensitivity. Evidence, polarity, and direction have smaller effects, mainly in biomedical predicates where these factors are explicit. Label-only scoring can produce a stronger scalar reaction when surface labels disappear, but it cannot say whether the damage concerns roles, direction, polarity, or evidence.

6.3 LLM factor judging as a complementary view

The LLM probe evaluates semantic recoverability rather than rule-based perturbation labels. Each judge saw the source edge, the factor inventory, and the perturbed text, then labeled each factor as covered, partial, or absent. Table 3 summarizes Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B.

Table 3. Exploratory LLM factor-judging probe. Judges are Gemini 2.0 Flash, GPT-4o-mini, and Llama-3.1-70B. Mean factor score maps covered, partial, and absent to 1, 0.5, and 0; agreements are computed only on overlapping factor judgments.

Probe	Items	Judges	Mean score	Pairwise agr.	Rule-label agr.
LLM factor judging	45	3	0.809	0.597	0.561

The mean LLM factor score is 0.809, pairwise agreement is 0.597, and agreement with rule-based perturbation labels is 0.561. These values indicate moderate stability rather than expert validation. Rule-based perturbations record intended damage; LLM judges estimate how much of the original factor remains recoverable from context. The two views are complementary: LEMON exposes controlled relation-level failures, while model-based judging shows which failures remain inferable.

7 Limitations and Error Sources

Two error sources recur in the experiments. The first is inventory incompleteness: predicates without decompositions, or with weak lexical cues, cannot be scored reliably. The error-driven expansion step exposes this issue before it contaminates the metric, but fallback decompositions remain less informative than expert-designed ones. The second is relation recoverability. WebNLG texts often repeat entity labels, so node recovery is high, while edge and fact recovery are substantially lower. The factor trace is the practical output in such cases: a low score can be traced to a predicate, a factor role, a missing cue, or weak subject/object evidence.

The study does not yet include expert human validation. The factor inventories are rule-based resources built from predicate labels, schema constraints, domain roles, lexical cues, and development-time candidates. The LLM probe adds a semantic recoverability check, but LLM judgments are not a replacement for expert annotation. Prior LLM-as-judge work shows that LLM judges can scale evaluation while introducing preference, calibration, and reasoning biases; the next version should add expert factor annotations and inter-annotator agreement [26, 27].

The evidence layer is limited. It can find labels and known cues, but it can miss implicit relations, long-range entailment, and paraphrases outside the cue dictionary. A stronger implementation should add embedding, entailment, or QA-based evidence while preserving the factor trace.

The reverse pipeline uses a lexical pseudo-extractor rather than a full text-to-KG system. This keeps the experiment controlled, but the reverse scores should be read as metric-behavior diagnostics. Similarly, the MINE-style baseline adapts KGGen’s node/edge information idea to graph-text pairs; it is not a reproduction of the original MINE benchmark.

LEMON also has a narrower scope than general semantic similarity. It assumes that a source graph, relation annotation, or predicate inventory is available. The present inventories cover roles, direction, polarity, causality, and ev-

idence where those dimensions are encoded by the predicate resources. They do not model pragmatic phenomena such as irony, implicature, presupposition, stance, or rhetorical force. The perturbations are auditable but template-like; they support construct-validity checks for relation deletion, argument swap, and polarity flip, but they do not prove robustness to natural paraphrase or state-of-the-art biomedical extraction performance.

8 Conclusion

LEMON evaluates KG–text fidelity by opening graph predicates into weighted semantic factors. It does not replace sentence-level metrics, triple matching, or MINE-style node/edge scoring. Its role is diagnostic: when a relation fails, it identifies whether the failure concerns relation class, roles, direction, polarity, or evidence.

The experiments support this distinction. On WebNLG, the same factor representation supports forward coverage and reverse recoverability. The MINE-style comparison shows that nodes are easier to recover than edges, and the relation-deletion check shows that entity-label coverage can remain almost unchanged while relation coverage drops. Perturbation tests show that LEMON reacts to the semantic dimensions represented in each inventory, with biomedical predicates making polarity, direction, causality, and evidence especially visible.

The component and LLM analyses clarify the result. Role factors contribute most to diagnostic sensitivity under controlled damage. Label-only scoring can be more severe as a scalar detector, but it cannot explain whether the damage concerns role assignment, direction, polarity, or evidence. LLM judges provide a complementary view of residual recoverability; they do not replace expert validation.

The main contribution is a layer between whole-triple factuality and sentence-level similarity: LEMON gives a reproducible factor trace for graph–text errors that coarser scores collapse into one number. Future work should validate the predicate inventories with domain experts, strengthen the evidence layer with paraphrase and entailment signals, and repeat the reverse experiment with stronger text-to-KG extractors.

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